

Supplementary Material for: Adaptive Neuromorphic Control of a Pendulum for Sustained Rhythmic Swing

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1. OVERVIEW

This supplementary material presents additional demonstrations and results supporting the manuscript.

2. CIRCUIT REPRESENTATION OF THE HCO MODEL

The neuromorphic control framework is governed by the state equations detailed in (S1). This system of nonlinear ordinary differential equations models the fast membrane potential $v_i(t)$, alongside the temporal evolution of the slow, $v_{s,i}(t)$, and ultra-slow, $v_{us,i}(t)$, state variables.

$$\begin{cases} \dot{v}_i(t) = \frac{1}{\tau_f} \left(-v_i(t) + g_f^- \tanh(v_i(t)) - g_s^+ \tanh(v_{s,i}(t)) \right. \\ \quad \left. + g_s^- \tanh(v_{s,i}(t) + 0.9) - g_{us}^+ \tanh(v_{us,i}(t) + 0.9) \right. \\ \quad \left. + I_{\text{syn},i}(t) + I_i(t) \right), \\ \dot{v}_{s,i}(t) = \frac{1}{\tau_s} (v_i(t) - v_{s,i}(t)), \\ \dot{v}_{us,i}(t) = \frac{1}{\tau_{us}} (v_i(t) - v_{us,i}(t)), \end{cases} \quad (\text{S1})$$

Figure S1 visualizes the corresponding circuit architecture for this model. The schematic maps the mathematical parameters to physical components, illustrating how the time constants ($\tau_f, \tau_s, \tau_{us}$) are realized via resistor-capacitor networks and how the synaptic gains ($g_f^-, g_s^+, g_s^-, g_{us}^+$) translate into amplifier configurations. This diagram provides a structural mapping of the internal feedback loops, self-excitation pathways, and reciprocal synaptic coupling, $I_{\text{syn},i}(t)$, defined in (S1).

3. VISUALIZATION OF PROPOSITION 1: NULLCLINE ANALYSIS OF SELF EXCITATION GAIN g_f^-

The system (S1) exhibits a fast-slow timescale separation where $\tau_f \ll \tau_s \ll \tau_{us}$. Thus, we can analyze the fast subsystem by treating the slow variables as quasi-static

on the fast timescale. Consequently, the fast dynamics are governed by $\dot{v}_i = \tau_f^{-1}(-v_i + g_f^- \tanh(v_i) + C)$, where the term $C = -g_s^+ \tanh(v_{s,i}) + g_s^- \tanh(v_{s,i} + 0.9) - g_{us}^+ \tanh(v_{us,i} + 0.9) + I_{\text{syn},i} + I_i$ acts as a slowly-varying bifurcation parameter encapsulating the network and slow-state feedback.

To isolate the structural conditions for bistability, which are dictated by the self-excitation gain g_f^- , we evaluate the nominal unperturbed case where $C = 0$. Because C enters the fast subsystem purely as an additive constant, non-zero values of C merely translate the fast nullcline vertically. This shifts the state-space coordinates of the equilibria but preserves the underlying geometric structure of the bifurcations parameterized by g_f^- . Setting $\dot{v}_i = 0$ and $C = 0$ yields the nominal equilibrium condition $v_i = g_f^- \tanh(v_i)$. Figure S2 illustrates the resulting saddle-node bifurcation characterized in Proposition 1 (in the main manuscript).

- (1) **Monostable Regime** ($g_f^- < 1$, blue): The maximum nullcline slope is less than one, yielding a single intersection that corresponds to a unique, globally asymptotically stable equilibrium.
- (2) **Bifurcation Boundary** ($g_f^- = 1$, magenta): The nullcline is tangent to the identity line at the origin, marking the critical threshold of a degenerate saddle-node bifurcation.
- (3) **Bistable Regime** ($g_f^- > 1$, red): The maximum slope exceeds one, generating three intersections. A central unstable equilibrium (saddle) separates two locally asymptotically stable outer equilibria, rendering the neural state bistable and allowing convergence to one of two non-zero activity levels.

4. VISUALIZATION OF PROPOSITION 2: SYNAPTIC GAIN $\{g_s^+, g_s^-, g_{us}^+\}$ MODULATION

In the HCO system (S1), the synaptic gains $\{g_s^+, g_s^-, g_{us}^+\}$ modulate the system dynamics by inducing vertical translations of the fast nullcline ($\mathcal{N} : \dot{v}_i = 0$). The direction

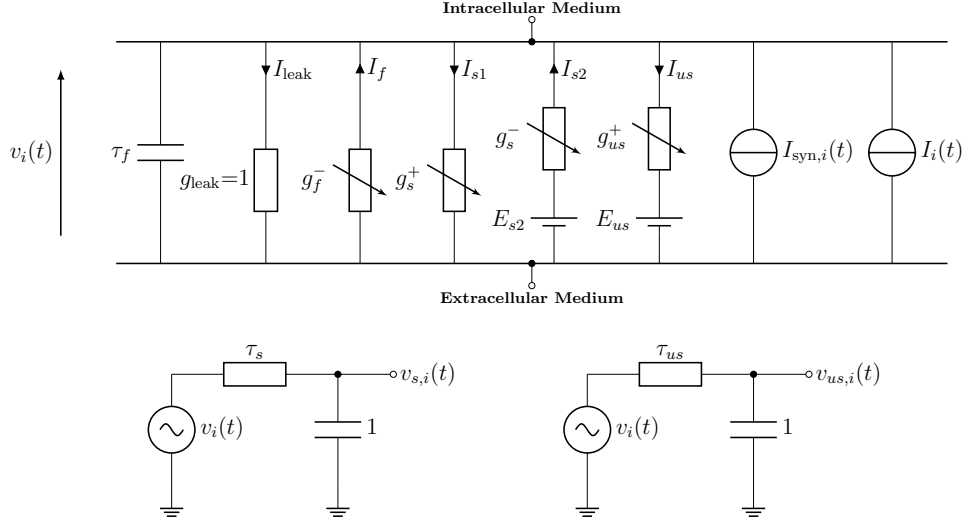


Fig. S1. Electrical circuit representing the Half-Center Oscillator (HCO) model defined in 1 in the main manuscript. The upper circuit models the membrane potential $v_i(t)$, where the membrane capacitance is defined directly by the time constant τ_f . The current arrows denote the physical direction of charge flow (inward or outward) to match the positive and negative signs in the mathematical formulation ($\tau_f \dot{v}_i = \sum I_{\text{in}} - \sum I_{\text{out}}$). Specifically, the outward (downward) currents are the passive decay $I_{\text{leak}} = v_i(t)$, the first slow current $I_{s1} = g_s^+ \tanh(v_{s,i}(t))$, and the ultra-slow current $I_{us} = g_{us}^+ \tanh(v_{us,i}(t) + 0.9)$. The inward (upward) currents are the fast current $I_f = g_f^- \tanh(v_i(t))$, the second slow current $I_{s2} = g_s^- \tanh(v_{s,i}(t) + 0.9)$, and the inputs $I_{\text{syn},i}(t)$ and $I_i(t)$. The voltage offsets inside the non-linear tanh functions act as reversal potentials, indicated by the batteries $E_{s2} = -0.9$ and $E_{us} = -0.9$. The lower sub-circuits illustrate the first-order RC low-pass filters that generate the state variables $v_{s,i}(t)$ and $v_{us,i}(t)$, normalized with a unit capacitance.

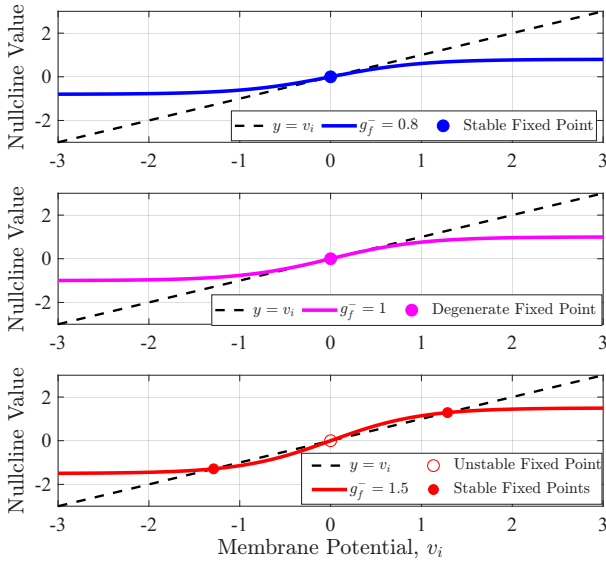


Fig. S2. The v_i -nullcline, $y = g_f^- \tanh(v_i)$, plotted for three self-excitation gain regimes: $g_f^- = 0.8 < 1$, $g_f^- = 1$, and $g_f^- = 1.5 > 1$. Fixed points correspond to the intersections of the nullcline $y = g_f^- \tanh(v_i)$ with the dashed line, $y = v_i$.

and magnitude of these translations are governed by the slowly varying term Φ , defined as

$$\Phi(v_{s,i}, v_{us,i}) = -g_s^+ \tanh(v_{s,i}) + g_s^- \tanh(v_{s,i} + 0.9) - g_{us}^+ \tanh(v_{us,i} + 0.9).$$

Figure S3 provides a geometric interpretation of Proposition 2 (see main manuscript) by plotting the identity line $y = v_i$ alongside the state-dependent fast nullcline curve $y = g_f^- \tanh(v_i) + \Phi(v_{s,i}, v_{us,i})$ as functions of the membrane potential v_i . The intersections of these two curves define the instantaneous equilibria of the fast subsystem. By vertically shifting this nonlinear nullcline curve, the slow dynamics orchestrate the requisite switching between active and quiescent states underlying limit cycle oscillations. This modulation occurs via three distinct mechanisms:

- (1) **Inhibitory Modulation** (g_s^+): Increasing the inhibitory gain from $g_s^+ = 0.5$ (blue) to 1.5 (red) forces a downward nullcline shift. This lowers the intersections with the identity line $y = v_i$, facilitating transitions to low-voltage equilibria and stabilizing the quiescent regime.
- (2) **Excitatory Modulation** (g_s^-): Conversely, increasing the excitatory gain from $g_s^- = 0.2$ (green dashed) to 1.5 (magenta dashed) drives an upward translation. This shift establishes and sustains high-voltage equilibria, driving the active bursting phase.
- (3) **Adaptive Modulation** (g_{us}^+): The shift from the cyan ($g_{us}^+ = 0$) to the black ($g_{us}^+ = 1.5$) curve illustrates time-dependent, ultra-slow adaptation. As the state $v_{us,i}(t)$ accumulates during an active phase, the nullcline progressively drifts downward. This drift ultimately destabilizes the high-voltage equilibria, acting as an autonomous burst-termination mechanism. The magnitude of g_{us}^+ scales this adaptation rate, directly dictating burst duration and oscillation frequency.

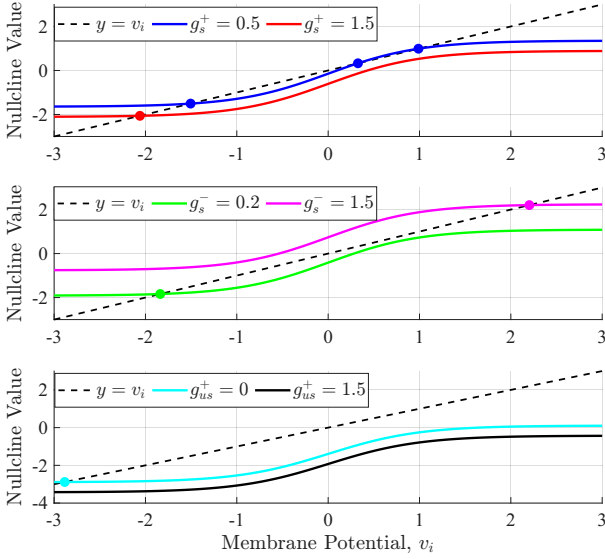


Fig. S3. Geometric influence of synaptic gains on the v_i -nullcline. The intersections between each nullcline and the identity line $y = v_i$ (black dashed line) define the system's fixed points, indicated by solid circular markers. Three modulation mechanisms are illustrated: (Top) variation of the inhibitory gain $g_s^+ \in \{0.5, 1.5\}$ (blue and red curves); (Middle) variation of the excitatory gain $g_s^- \in \{0.2, 1.5\}$ (green and magenta curves); and (Bottom) evolution of the ultraslow variable $g_{us}^+ \in \{0, 1.5\}$ (cyan and black curves), highlighting adaptive, time-dependent modulation of the nullcline geometry. The self-excitation parameter is fixed at $g_f^- = 1.5$, while all remaining gains are held constant within each respective scenario.

These systematic nullcline transformations govern the existence and stability of the fast subsystem equilibria, ultimately driving the rhythmic switching behavior of the neural oscillator.

5. VISUALIZATION OF PROPOSITION 3: BIFURCATION CONTROL VIA EXTERNAL INPUT $I_i(t)$

Supplementary Figure S4 provides a geometric interpretation of Proposition 3 (see main manuscript), illustrating how the external input current $I_i(t)$ functions as a bifurcation control parameter. To visualize this mechanism, the figure plots the identity line $y = v_i$ alongside the input-modulated nonlinear curve $y = g_f^- \tanh(v_i) + I_i$ as functions of the membrane potential v_i . Setting $\dot{v}_i = 0$, the intersections of these two curves define the instantaneous equilibria of the fast subsystem.

In the unforced baseline state ($I_i \approx 0$, blue curve), the system operates in a bistable regime ($g_f^- > 1$). This is characterized by three distinct equilibria: two locally asymptotically stable nodes located on the high- and low-voltage branches, separated by a central unstable saddle point.

The application of a positive step in the input current ($I_i = 1$, red curve) acts purely as an additive term, inducing a strict upward vertical translation of the nonlinear

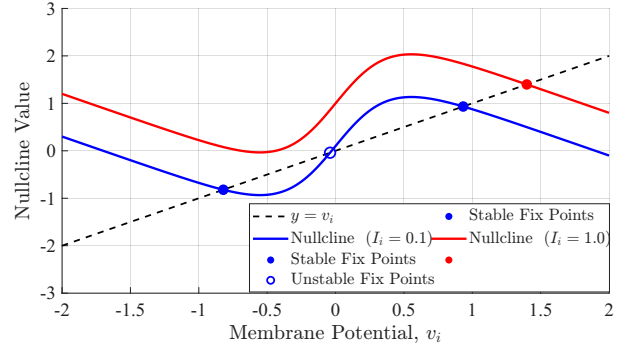


Fig. S4. Geometric effect of the external input current I_i on the fast subsystem equilibria. The equilibria correspond to the intersections of the identity line $y = v_i$ (dashed black) and the input-dependent curve $y = g_f^- \tanh(v_i) + I_i$. The unforced state ($I_i \approx 0$, blue line) exhibits bistability. Applying a positive external input ($I_i = 1$, red line) produces an upward vertical translation that eliminates the low-voltage resting state via a saddle-node bifurcation. Equilibria are indicated by circles (filled for stable, open for unstable).

nullcline. This translation triggers a saddle-node bifurcation that annihilates the low-voltage stable equilibrium and the unstable saddle.

Practically, this geometric transformation represents the mechanism of external state switching. By eliminating the low-voltage equilibrium, the positive input destabilizes the quiescent resting state, compelling the system's trajectory to converge to the unique remaining stable equilibrium on the upper branch. This defines how external stimuli or network commands force a quiescent neuron into an active, high-voltage bursting regime.

6. EXTERNAL INPUT $I_i(t)$ AS AN EVENT-TRIGGERED FEEDBACK MECHANISM

As established in Proposition 3 (see main manuscript) and Section 5, the external input current $I_i(t)$ acts as a bifurcation parameter that directly modulates the excitation threshold of the fast subsystem. This property can be systematically exploited to achieve phase tuning via closed-loop feedback.

In the primary case study of the manuscript, $I_i(t)$ is maintained at a constant value to demonstrate the intrinsic limit-cycle properties of the HCO. However, Proposition 3 implies that this constant bias can be replaced with sparse, event-triggered phase correction pulses derived from plant feedback. Under this paradigm, the controller does not forcefully impose a predetermined reference frequency. Instead, the oscillator entrains to the natural resonance of the mechanical plant. By detecting specific physical events—such as positional extrema or velocity zero-crossings—the controller injects sparse energy pulses via $I_i(t)$ at optimal phases, allowing the plant's natural frequency to govern the closed-loop rhythm. This establishes a highly efficient, resonance-driven control cycle: physical plant event $\rightarrow$ sparse feedback pulse via $I_i(t)$ $\rightarrow$ HCO phase entrainment $\rightarrow$ mechanical actuation.

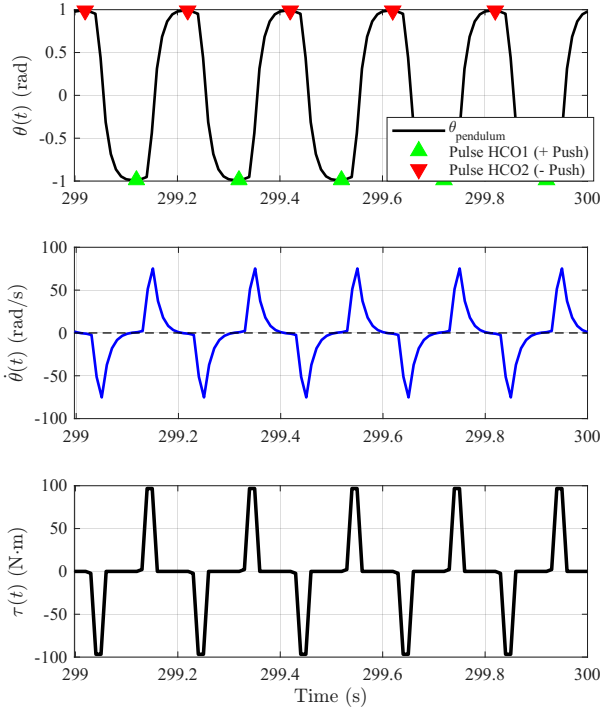


Fig. S5. Steady-state closed-loop dynamics ($t = 299$ to 300 s). (Top) Pendulum angle θ . (Middle) Angular velocity $\dot{\theta}$. (Bottom) Event-triggered feedback modulation injected precisely at the mechanical extrema ($\dot{\theta} = 0$) to dynamically synchronize the oscillator phase with the physical plant.

Supplementary Figure S5 illustrates the efficacy of this event-triggered feedback strategy. By utilizing the plant states to pulse $I_i(t)$, the HCO is synchronized to generate positive torque precisely at the minimum angular displacement (where θ is at a minimum and $\dot{\theta} = 0$) and negative torque at the maximum angular displacement (where θ is at a maximum and $\dot{\theta} = 0$). This phase alignment enables actuating the system exactly at the kinematic extrema of the pendulum's swing.

7. CONCLUSION

This supplementary material provides visualizations and analyses supporting Propositions 3 of the main manuscript. By illustrating the effects of the self-excitation gain (g_f^-), synaptic feedback gains (g_s^+, g_s^-, g_{us}^+), and the external input current ($I_i(t)$) on the fast subsystem nullclines, we establish a foundation for the analysis and design of HCO-driven control inputs. Furthermore, we demonstrate that while $I_i(t)$ can be leveraged as a closed-loop, event-triggered feedback channel to achieve dynamic phase entrainment with a physical plant. The source code and dataset supporting the findings of this study are publicly available.¹

¹ The MATLAB code and simulation data reproducing the results in this study are publicly available at [GitHub](#).